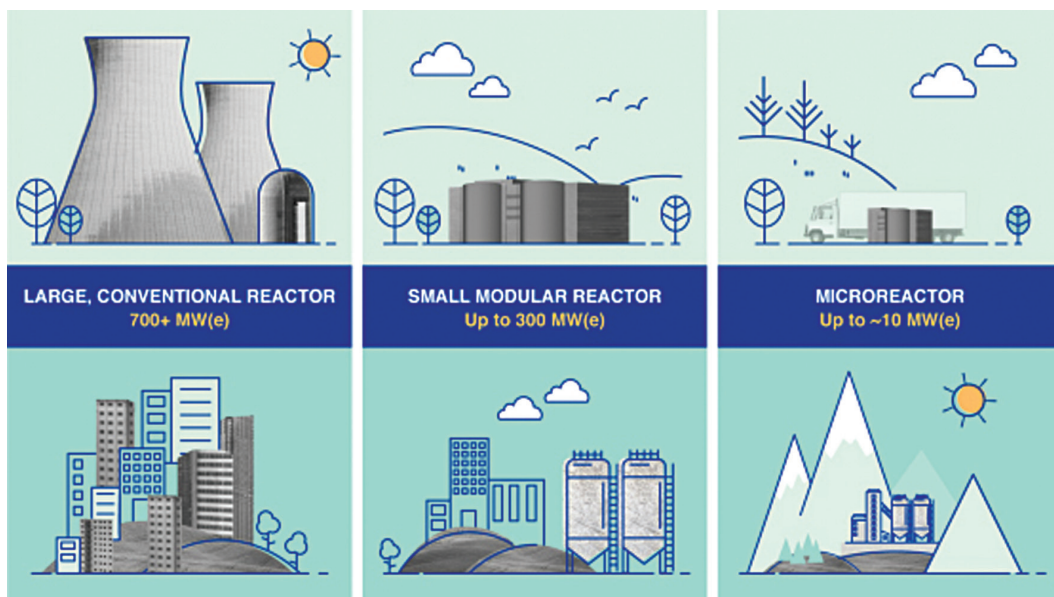


SMALL MODULAR REACTORS For Sustainable Energy Transition

SMRs have shifted from science fiction to ready science. To achieve clean, stable, scalable energy, it's time to move beyond legacy reactors and invest in this compact revolution.



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Classification of reactors (Source: <https://www.iaea.org>)

“The potential of SMRs is immense, providing scalable, secure, and environmentally responsible energy solutions that the world needs.”

—International Energy Agency (IEA)

Nuclear power is set to play a pivotal role in the clean energy transition, offering high-density, carbon-free, and reliable energy that complements the variability of renewables. Though not renewable, nuclear energy generates substantial power using minimal land and operates independently of weather conditions.

Small modular reactors (SMRs) mark a major advancement in nuclear technology. ‘Small’ refers to their power output; ‘modular’ to their unit-based assembly; and ‘reactor’ to the system where controlled nuclear fission

occurs. These compact, factory-built units typically generate 300MWe of power, equivalent to 7.2 million kWh per day. By comparison, large nuclear reactors produce 1000MWe, or 24 million kWh daily. SMRs are engineered to deliver flexible, scalable, and inherently safe energy solutions.

SMRs have a capacity factor exceeding 90%, making them a dependable baseload power source compared to intermittent renewable sources. Modular construction, scalability, rapid on-site assembly, and a reduced emergency planning zone (EPZ) (see Box 1) make them well-suited for deployment in remote areas, industrial hubs, urban centres, and off-grid locations. They have versatile applications, including desalination, district heating, and hydrogen production.